E-commerce sales data using Excel

Project Objective:To analyse E-commerce sales data using Total_Amount(Revenue)descriptive statistics and pivot tables to find performing products,region,Month-wise sale and store-type.to provide business insights through 4 types of analytic.
Dataset Description: E-commerce (Sales-fact)sales cleaned data dataset with 2000 records, containg orderdate,product category,store-type,Region,Total_amount

Column Description

Column Name	Description
Order_date	Date of Purchase(jan-Dec2024)
Product_Category	Types of product - beauty,clothing,Elctronics,home , sports.
Store_type	Flagship, Online, Outline
Store_Region	East, North, South ,West
Total _Amount	Final sales amount in rupees
Product_cost	Cost price of product
Customer Id	Unique coustomer id

Data cleaning & transformation:

Formula used:

With screenshot

*total cost=product cost *quantity

*profit=Total Amount-Total cost

loss=profit/total amount*100

*using VLOOKUP(D2,'Product_cleaned data'!A:F,3,FALSE)to fetch product category

*using =VLOOKUP(E2,Store_Dim!A:E,5,FALSE)to detch store tye from store_dim

*using =VLOOKUP(E3,Store_Dim!A:D,3,FALSE)to fetch sore region

Group order date by months for month_wisereport

Created pivot table for category,Region, store-type,moth-wise report

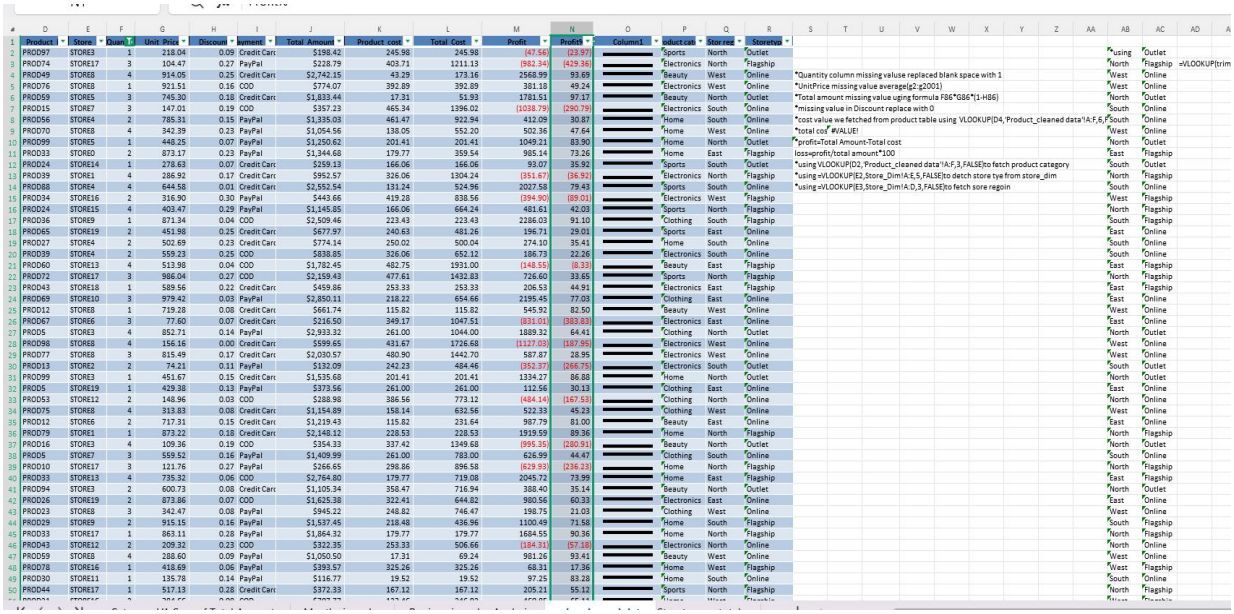
*product id replaced p- with prod

*cost missing value average(f2:f101)

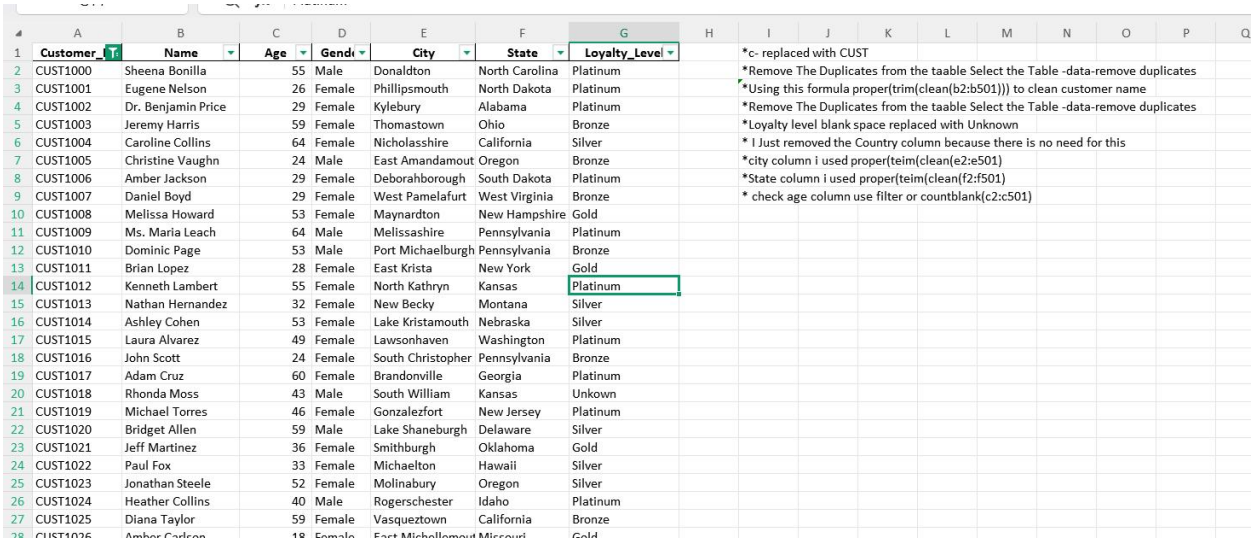
*stock column missing value

average(g2:g501)

Sales cleaned date



| Product | Store | Quan | Unit Price | Discount | ayment | Total Amount | Product cost | Total Cost | Profit | Profit | Column1 | oduct cat | Stor reg | Storetyp | | |



Customer_ID	Name	Age	Gender	City	State	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	Platinum
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamout	Oregon	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	Unkown
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	Bronze

*c- replaced with CUST
*Remove The Duplicates from the taable Select the Table -data-remove duplicates
*Using this formula proper(trim(clean(b2:b501))) to clean customer name
*Remove The Duplicates from the taable Select the Table -data-remove duplicates
*Loyalty level blank space replaced with Unknown
* I Just removed the Country column because there is no need for this
*city column i used proper(teim(clean(e2:e501)
*State column i used proper(teim(clean(f2:f501)
* check age column use filter or countblank(c2:c501)

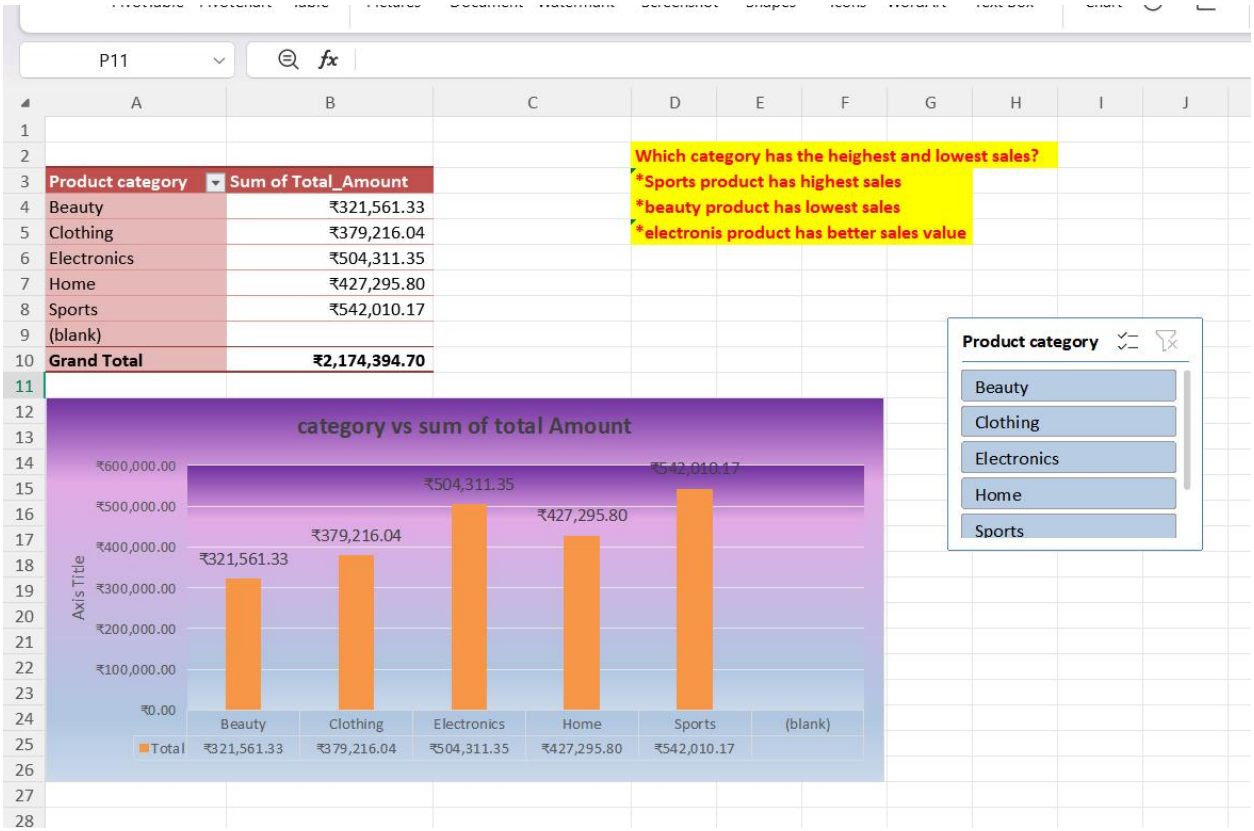
Product_clean table



Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	254.62	352
PROD3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261.00	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.80	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.40	273
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	245.78	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	272
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	484.62	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	248.82	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340
PROD25	Women's Scarf Set	Clothing	Accessories	Brown and Sons	174.02	268

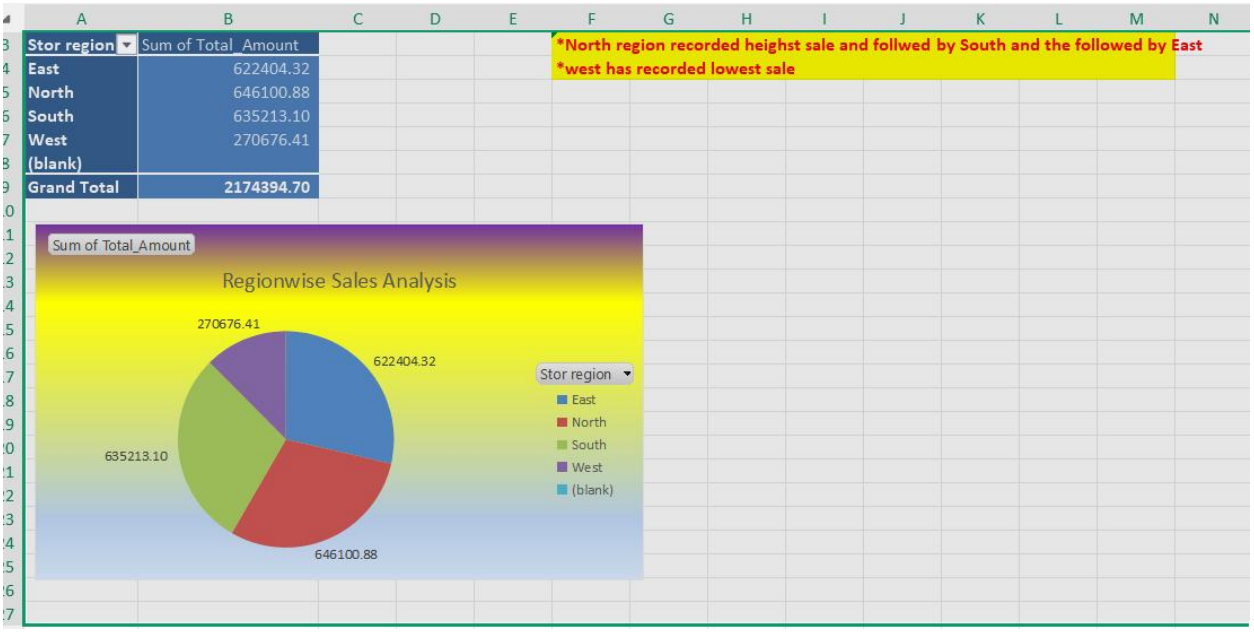
*product id replaced p- with prod
*cost missing value average(f2:f101)
*stock column missing value average(g2:g501)

Category vs Sum of Total_amount



Product category	Sum of Total_Amount
Beauty	₹321,561.33
Clothing	₹379,216.04
Electronics	₹504,311.35
Home	₹427,295.80
Sports	₹542,010.17
(blank)	
Grand Total	**₹2,174,394.70**

Regional wise Report



Stor region	Sum of Total_Amount
East	622404.32
North	646100.88
South	635213.10
West	270676.41
(blank)	
Grand Total	**2174394.70**

Monthwise_Sales repot

Order_Date	Sum of Total_Amount
Jan	91986.10
Feb	72055.66
Mar	107588.86
Apr	102393.30
May	88686.74
Jun	65636.01
Jul	101129.55
Aug	99833.22
Sep	88892.92
Oct	87118.63
Nov	102983.30
Dec	88877.87
Grand Total	**1097182.16**

Which month has highest and lowest ?

*HIGHEST SALES WERE RECORDED IN MARCH AND FOLLOWED BY NOVEMBER

PEAK SALES OBSERVED IN MARCH AND NOVEMBER

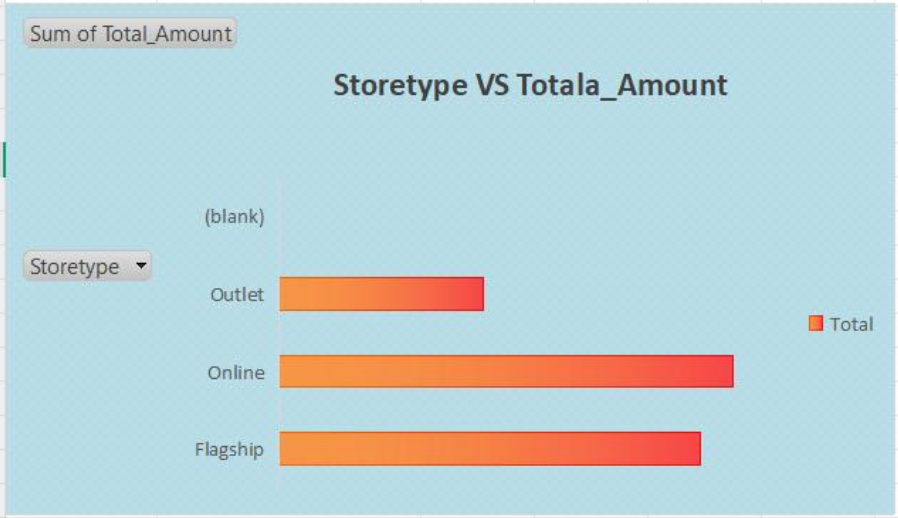
*June recorded the lowest salse



Store_type vs Sum of Total_amount

Storetype	Sum of Total_Amount
Flagship	848576.67
Online	914253.52
Outlet	411564.51
(blank)	
Grand Total	2174394.70

Whichstore type generates highest revenue?
***Online purchase recrded the heighest sales**
*** Customers prefer online shopping than other two**



Descriptive Statistics

Summary Statistics			
Sum(Total)	$2,174,394.70		
Mean(Average)	1087.20		Sum(g2:g2001)
Mode	#N/A		average(g1:g2001)
Median	868.72		MEDIAN(G2:G2001)
Varience	671439.36	1128430.6	mode(g2:g2001)
Standard deviation	819.41	1062.28	stdev.p(g2:g2001)
			STDEV.S(G3,G8,G15,G21,G24)
			var.p(g2:g2001)
Min	18.46		var.s(g3,g8,g15,g21,g24)
Max	3908.24		min(g2:g2001)
Count	2000		max(g2:g2001)
			count(g2:2001)

Statistical Summary Using Tool Pack	
Total_Amount	
Mean	1087.197345
Standard Error	18.32723662
Median	868.72
Mode	1106.31
Standard Deviation	819.6189383
Sample Variance	671775.2041
Kurtosis	0.045826542
Skewness	0.879047022 positively skewed
Range	3889.78
Minimum	18.46
Maximum	3908.24
Sum	2174394.69
Count	2000
Largest(1)	3908.24
Smallest(1)	18.46
Confidence Level(95%	35.9424852

198.42	245.98	-639.9957365	245.98				
228.79	1211.13		403.71				
2742.15	173.16		43.29				
774.07	392.89		392.89				
1833.44	51.93		17.31				
357.23	1396.02		465.34				
1335.03	922.94		461.47				
1054.56	552.2		138.05				
1250.62	201.41		201.41				
1344.68	359.54		179.77				
259.13	166.06		166.06				
952.57	1304.24		326.06				
2552.54	524.96		131.24				
443.66	838.56		419.28				
1145.85	664.24		166.06				
2509.46	223.43		223.43				
677.97	481.26		240.63				
774.14	500.0431429		250.0215714				
838.85	652.12		326.06				
1782.45	1931		482.75				
2159.43	1432.83		477.61				
459.86	253.33		253.33				
2850.11	654.66		218.22				
661.74	115.82		115.82				
216.50	1047.51		349.17				
2933.32	1044		261				
599.65	1726.68		431.67				
2030.57	1442.7		480.9				
132.09	484.46		242.23				
1535.68	201.41		201.41				
373.56	261		261				
288.98	773.12		386.56				
1154.89	632.56		158.14				
1219.43	231.64		115.82				

Visualization and Insights Arrived with screenshot of visualization

Report/Dashboard screenshot:

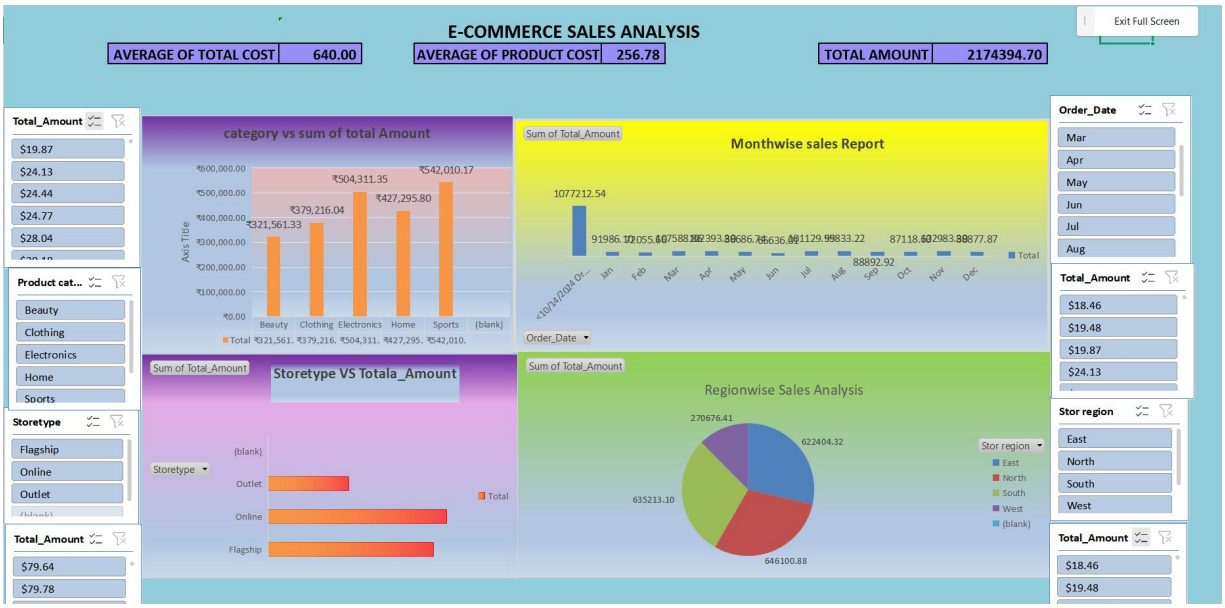
- **<u>Descriptive analysis:</u>**
- **Which category has the highest and lowest sales?**
- Sports product has highest sales
- Beauty product has lowest sales
- Electronics have better sale value
    - **Which month has highest and lowest sale?**
    - Highest sales were recorded in march and followed by November peak sale is recorded in march and November
    - Lowest sale in june month
    - **Which region perform best?**
    - North region recorded highest and followed by south and then East.
    - West has recorded lowest sale.
    - **Which store type generates highest revenue?**
    - Online purchase recorded the highest
    - Customer like online purchase other than two.
- **<u>Diagnostic Analysis:</u>**
- Why sports is highest means sports product have a good balance of price and high demand due to fitness trends.
- Beauty is high quantity but low price
- **Why north is highest and west is lowest?**
- North region has more Flagship stores and a larger customer base,
- West region has on;y outlet store which sells low-cost items.
- Customer prefers comfort and offers, online shopping gives discount and home delivery,
- March is financial year -end and November is festival like Diwali, puja holidays .June is low due to summer when shopping is less .

- **<u>Predictive Analysis:</u>**
- If the trends constitutes we can expect highest sale on next March and November. June will be again low.

- Category : People are more fascinated in sports nowadays so sports will grow next year too.we should provide more discount on beauty products then we can increase the beauty product .
- Regional: If we can't take any action, west will drop from current revenue so focus on the west part. North will increase.

- **Prescriptive Analysis:**
- Sports and electronics products going good. But we need to focus on beauty products that we offer some combo offers like buy sports get beauty it will lift beauty items go up.
- To make online sales in the west .otherwise we can't improve the sale, To increase summer sale in June like March an November offer stratergy .

Dashboard Image:



4 type of insights as storytelling:

Total 21,74,394 from 2000 orders.sports top beauty low.

*North top region west lowest region.

*online shopping has highest sales other than two.

Conclusion: our E-commerce business is profitable with 21,74,394L revenue from 2000 orders.Average key sales are sports category,North Region and Online purchase.